

Security Risk Assessment Report

Introduction

This report outlines key security risks that can affect an organization's network and systems. It also provides practical recommendations to reduce these risks and improve overall cybersecurity. The focus is on three important security hardening methods: strong password policies, firewall configuration, and multifactor authentication (MFA). These measures help protect company data, prevent unauthorized access, and reduce the likelihood of cyberattacks.

Part 1: Recommended Hardening Tools and Methods

1. Strong Password Policies and Management

Strong password policies help protect user accounts from unauthorized access. Many cyberattacks occur because employees use weak or shared passwords that attackers can easily guess or steal.

Recommended Actions

- Require employees to create unique and complex passwords.
- Prevent password sharing between staff members.
- Enforce regular password updates.
- Block the reuse of old passwords.
- Encourage the use of password managers for secure storage.

Benefits

Implementing strong password policies reduces the risk of:

- Brute-force attacks
 - Credential stuffing attacks
 - Unauthorized account access
 - Internal misuse of accounts
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2. Firewall Configuration

A firewall acts as a security barrier between the company's internal network and external networks such as the internet. Proper firewall configuration helps monitor and control incoming and outgoing traffic.

Recommended Actions

- Configure firewall rules to allow only approved traffic.
- Block suspicious or unauthorized access attempts.
- Restrict unnecessary services and ports.
- Monitor firewall logs regularly for unusual activity.

Benefits

Proper firewall configuration helps:

- Prevent hackers from accessing internal systems
 - Reduce malware and ransomware threats
 - Protect sensitive business information
 - Improve overall network security
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3. Multifactor Authentication (MFA)

Multifactor Authentication (MFA) adds an extra layer of security by requiring users to verify their identity using more than one method.

Examples include:

- Password + mobile verification code
- Password + fingerprint scan
- Password + authentication app

Recommended Actions

- Enable MFA for all employee accounts.
- Prioritize MFA for administrator and remote-access accounts.
- Use secure authentication applications where possible.

Benefits

MFA significantly reduces the risk of:

- Stolen password attacks
- Unauthorized login attempts
- Account compromise

- Data breaches

Even if a password is stolen, attackers still cannot access the account without the second verification factor.

Part 2: Explanation of Recommendations

Recommendation 1: Enforce Strong Password Policies

Why It Is Important

Weak passwords are one of the most common causes of security breaches. Attackers often use automated tools to guess passwords or reuse stolen credentials from previous breaches.

Effectiveness

Strong and unique passwords make it much harder for attackers to gain access to systems. Regular password changes also reduce the chances of long-term unauthorized access.

Implementation Frequency

- Password policies should be active at all times.
 - Password audits should be performed regularly.
 - Passwords should be changed every 60–90 days.
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Recommendation 2: Configure Firewalls with Proper Rules

Why It Is Important

Without a properly configured firewall, attackers may gain unauthorized access to company systems and sensitive information.

Effectiveness

A firewall filters harmful traffic and only allows approved communication into the network. This reduces exposure to cyber threats and helps protect internal systems.

Implementation Frequency

- Firewall rules should be configured immediately.
- Reviews should occur every quarter.
- Rules should also be updated after any major network or system change.

Recommendation 3: Implement Multifactor Authentication (MFA)

Why It Is Important

Passwords alone are no longer enough to secure accounts because they can be stolen through phishing, malware, or data breaches.

Effectiveness

MFA provides an additional security layer that greatly reduces the chance of unauthorized access. Even if attackers know the password, they still need the second authentication factor.

Implementation Frequency

- MFA should remain permanently enabled across all systems.
- Monitoring should be continuous to ensure compliance and effectiveness.
- MFA settings should be reviewed regularly.

Conclusion

Implementing strong password policies, properly configured firewalls, and multifactor authentication will significantly improve the organization's cybersecurity posture. These measures work together to reduce vulnerabilities, protect sensitive information, and defend against common cyber threats. Regular monitoring, updates, and reviews are essential to ensure that these security controls remain effective over time.